

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence COURSE CODE: CSA1707

COURSE OUTCOME COVERED

CO5: Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using

scalable systems and integrate components in distributed AI applications. (BTL 5)

Assessment Tool Weightage: 50%

Duration : 1 Hour

QUESTIONS: 5 Total Marks:50

Annexure A

Reverse Engineering

Code of Conduct:

I J O JOANNA DIVINE(192572003) (Name / Reg No) certify that this submission is my original work and that I have adhered

to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Joanna

Below is a **similar Annexure A format** for a **Reverse Engineering** assignment, with clear descriptions, student activities, and expected outcomes.

Annexure A – Reverse Engineering

1. Machine Learning Techniques for Reverse Engineering

Description:

Machine Learning techniques can be applied to reverse engineering to classify software components, identify programming patterns, detect anomalies, and predict the functionality of unknown code. Students explore supervised, unsupervised, and semi-supervised learning approaches for analyzing software.

Student Activity:

- Collect a dataset of source-code samples.
- Extract relevant code features.
- Apply classification or clustering algorithms.
- Analyze similarities and differences among programs.
- Compare the performance of different ML algorithms.

Expected Outcome:

Students will evaluate the suitability and effectiveness of different ML algorithms for software reverse-engineering tasks.

2. AI for Source Code Analysis and Program Understanding

Description:

AI techniques can analyze large source-code repositories and automatically identify functions, classes, variables, dependencies, and programming patterns. NLP and Large Language Models can also generate human-readable explanations of complex source code.

Student Activity:

- Select a suitable open-source repository.
- Analyze the source code using AI-assisted tools.
- Identify important functions, classes, and dependencies.
- Generate natural-language explanations of selected code.
- Verify the AI-generated explanations against the actual implementation.

Expected Outcome:

Students will assess the effectiveness, accuracy, and limitations of AI techniques in understanding unfamiliar source code.

3. AI-Based Malware Reverse Engineering and Analysis

Description:

AI can assist security researchers in analyzing malware by identifying suspicious code patterns, API calls, strings, functions, and behavioral indicators. Students study the conceptual workflow of AI-assisted malware analysis using safe, non-executable datasets or publicly available research datasets.

Student Activity:

- Study publicly available malware-analysis datasets.
- Identify relevant static features.
- Apply suitable ML classification techniques.
- Analyze suspicious patterns and indicators.
- Evaluate classification accuracy using appropriate metrics.

Expected Outcome:

Students will understand how AI can support automated malware analysis and evaluate the performance of ML-based malware classification techniques.

4. Reverse Engineering of APIs Using AI

Description:

When API documentation is incomplete or unavailable, AI can help analyze requests, responses, source code, and observed behavior to infer endpoints, parameters, data formats, and relationships between API components.

Student Activity:

- Select a documented open API or controlled test API.
- Analyze its request-response structure.
- Use AI techniques to infer API functionality.
- Generate API documentation from the analysis.
- Compare the generated documentation with the original API specification.

Expected Outcome:

Students will assess the usefulness of AI for API reverse engineering, functionality discovery, and automated documentation generation.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness